Pranav Rathod

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Education

University of Southern California — Los Angeles, CA
Master of Science in Computer Science - Multimedia and Creative Technologies — *GPA: 3.60/4.00*

University of Illinois at Chicago — Chicago, IL
Bachelor of Science in Computer Science, Minor in Art — *GPA: 3.92/4.00*

Relevant Coursework

Multimedia Systems Design, Analysis of Algorithms, Computer Graphics, Computer Animation and Simulation, Game Engine Development, 3D Graphics and Rendering, Augmented/Virtual Reality, Artificial Intelligence, Framework-based Development

Skills

Languages: C/C++, Java, Python, C#, Dart, Go, SQL, F#, JavaScript
Technical: OpenGL, PyTorch, WebGL, Unity, Vuforia Engine, Blender, Flutter SDK, GCP, Firebase, SQLite3, Multi-Threading, Locking, TCP/IP, Android Studio, Google/JUnit5 Testing Framework, Docker, Git, MacOS, Windows, Unix/Linux,

Experience

Media Support Specialist — January 2024 – Present
Annenberg TechOps Information Technology, University of Southern California — *Los Angeles, CA*
- Leveraging web and design expertise and providing coding support to make custom news feed and article pages.
- Answering technical questions and provided assistance with HTML, Adobe Creative Cloud and WordPress.
- Producing screen casts and step-by-step tutorials to aid users in multimedia production.

Research Assistant — June 2022 – December 2023
ELiCIT Lab, University of Illinois at Chicago — *Chicago, IL*
- Delivered Flutter app for Natural-User Interaction studies, utilizing GCP's Speech-to-Text, Meta's Detectron2 and PyTorch.
- Investigated student interactions on PufferTouch display using Unity and C#, studying its impact on collaborative learning.
- Collaborated with a multidisciplinary team of 5 researchers to integrate user feedback into UI enhancements.
- Explored existing research on creative technologies for children, contributing to a comprehensive literature review.

Teaching Assistant — January 2022 – December 2022
Department of Computer Science, University of Illinois at Chicago — *Chicago, IL*
- Guided students in debugging and writing code in C, C++ and building applications using Flutter, Dart across 3 semesters.
- Formulated course logistics and graded assignments for intermediate and advanced Computer Science courses each semester.
- Utilized communication platforms such as Piazza and Blackboard to effectively instruct more than 100 students at once.

Projects

Segmented Video Compression | *Python, Detectron2, OpenCV, PyTorch* — December 2024
- Designed a compression pipeline to encode .rgb videos using DCT, quantization, and motion estimation.
- Advanced segmentation by using Detectron2 and optical flow, achieving 95% accuracy in foreground-background separation.
- Constructed a video player enabling 30 FPS playback with real-time IDCT reconstruction, and synchronized audio.

Roller Coaster Simulation | *C++, OpenGL* — March 2024
- Engineered a roller coaster simulation with Catmull-Rom splines in OpenGL, using control points from a custom format file.
- Elevated appearance with rail cross-section rendering and introduced a texture-mapped ground with Phong shading.
- Automated frame capture for animation with an immersive first-person view and realistic camera movement.

Virtual Reality Kiosk | *Unity, C#, Virtual Reality ToolKit (VRTK), Blender* — October 2023
- Developed a Virtual Reality environment showcasing the interior of a proposed college building.
- Allowed users to move and interact with objects by deploying application in a VR headset.
- Built custom 3D models leveraging Blender and mapped custom textures drawn using Procreate.

Shadow Maps | *HTML5, JavaScript, WebGL* — April 2023
- Implemented WebGL shadow mapping for realistic rendering of shadows in urban settings.
- Developed interactive WebGL app with file upload, configurable projection, and lighting controls.
- Optimized performance with framebuffer objects, performed advanced features like percentage closer filtering.

Awards

Undergraduate Research Forum | *Chicago, IL* — April 2023
- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research.
- Presented research findings and methodologies, effectively communicating complex technical concepts to a diverse audience.

Engineering Expo | *Chicago, IL* — April 2021
- Won 'Best in Category' for designing an Arduino-based device to enforce COVID-19 social distancing.
- Tracked the number of people in an enclosed space, along with the room's temperature.